PERSONAL AIR EXPOSURE COMPANION

MVP Specification & Developer Build Guide

Understand where your day meets polluted air — and find practical ways to reduce avoidable exposure.

Product type Personal air-exposure tracking companion	**Initial audience** Students, office workers and daily commuters in Indian cities
Release goal A useful, honest, working MVP	**Document purpose** Product requirements for a developer or hackathon team

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Important: this product estimates exposure from available environmental data and user-entered routines. It does not measure exactly what a person inhales and is not a medical device.

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How to use this document: Treat it as a build specification, not application code. Items marked “verify” must be checked against current provider documentation and the hackathon's official rules before implementation.

01. Executive Summary

The Personal Air Exposure Companion helps people understand the pollution they may encounter across a normal day. A city AQI value is useful, but it cannot tell a person whether their own routine involved a busy road, an outdoor walk, a filtered room, or a long commute.

Users choose their city, save common places, and enter activities with start times or durations. The app retrieves available PM2.5 data or clearly labelled forecasts for relevant areas. It combines those values with the user's time at each place to produce an estimated daily timeline and a time-weighted average concentration.

The app highlights the parts of the routine that contribute most to the estimate and suggests practical actions, such as checking air quality before outdoor exercise or choosing a less busy time for a walk. It also shows data source, measurement time, coverage and uncertainty. It must never pretend that a nearby station is a personal sensor.

Initial users: students, office workers and daily commuters in cities such as Delhi, Kanpur and Mumbai. The first release should focus on one dependable data path, manual activity entry, transparent calculations and a clean demo—not AI, complex tracking or many cities at once.

Product promise: “See the pollution your routine may expose you to, understand the estimate, and make a better-informed next choice.”

02. Problem Statement

Most AQI apps report an area-level condition at a particular time. AQI is an index made from pollutant readings; it is not itself a concentration that can be multiplied by hours to calculate exposure. A city-wide or station reading also may not match the air at every street, indoor room or route.

Two people can live in the same city and see the same city AQI, yet have different routines. Person A works from home, spends most of the day indoors and takes a short walk in a park. Person B commutes beside heavy traffic, waits outdoors and exercises next to a main road. Their potential exposure patterns may differ, even if both check the same city AQI.

The product addresses the gap between **“What is the air like around the city?”** and **“What does my own day look like, based on the information available?”** It cannot know the exact air a person breathed unless suitable personal measurement devices are used. The MVP is an estimate, with visible limits.

03. Target Users

User group	Typical problem	What they need from the MVP
Students	Commute, campus time, outdoor sports and changing schedules.	Fast manual activity logging; a simple daily view; practical, non-alarmist tips.
Office workers	Travel at rush hour; time split between home, office and outdoors.	Saved places; commute blocks; weekly pattern comparison.
Daily commuters	Exposure can vary by route, traffic and time of day.	A clear view of which logged periods have higher estimated concentration.
Hackathon judges / early testers	Need to see a credible problem and a working, understandable solution.	Transparent data sources, a reproducible calculation and an end-to-end demo.

MVP expectations: simple setup, useful results without a wearable, no forced background tracking, and an honest explanation whenever data is missing or approximate.

04. Product Vision

The vision is to move from a city-only snapshot to a personal daily story, using user-entered routine plus nearby environmental information.

A memorable product statement: **“Your day, your places, your estimated exposure.”**

How it differs from a basic AQI app: it connects place + time + available PM2.5 concentration; shows where the estimate is coming from; and suggests actions tied to the user's logged routine. It does not claim to calculate a precise dose or diagnose health risk.

05. User Journey

First-time user flow:

- 1. Open the app and read a short note explaining that exposure is estimated, not directly measured.
- 2. Select a city (for example, Kanpur) and optionally allow location access. Manual city/place entry must remain available.
- 3. Add common places such as Home, College and Bus Stop. Use a map pin or a typed location if supported.
- 4. Enter today's activities: College, 09:00–15:00; commute, 08:00–09:00; outdoor walk, 17:00–17:30.
- 5. The backend finds suitable recent observations or a clearly labelled forecast for each place/time. It records the source and timestamp.
- 6. The app checks for missing, old or invalid data, then estimates a time-weighted average for the tracked periods.
- 7. The Dashboard shows the day's timeline, estimate, coverage and the periods contributing most to the estimate.
- 8. The user opens a recommendation, sees why it was suggested, and can compare weekly patterns as more days are logged.

A useful result should still be possible when data is incomplete: show the tracked time and gaps, rather than silently filling them with invented measurements.

06. MVP Features

Each feature below includes the first-release boundary. Build the smallest version that makes the core promise understandable and testable.

6.1 Simple onboarding and city selection

- **What:** Choose a city and preferred units/display options; explain the estimate and ask for optional location permission.
- **Why:** A user should understand the app and be able to start without giving precise location.
- **Interaction/example:** User selects Kanpur, types 'College' and continues without enabling GPS.
- **MVP:** City selection, privacy note, optional permission and manual fallback. **Later:** account sync and onboarding personalization.

6.2 Adding frequently visited places

- **What:** Save a place name and a location reference (coordinates or supported geocoding result).
- **Why:** Users repeat places and should not re-enter them each day.
- **Example:** Home, HBTU campus, bus stop. Display the source area used for the estimate.
- **MVP:** Add/edit/delete places and use manual place entry. **Later:** automatic place detection and route mapping.

6.3 Manual entry of daily activities and time

- **What:** Log activity, place, start/end time or duration.
- **Why:** Time is required to weight concentration values; activity type alone is not a measurement.
- **Example:** 'Commute — 08:00 to 08:45 — Bus Stop to College' (MVP may store a single representative place for this block).

- **MVP:** Add, edit and delete time blocks; validate overlapping or impossible times. **Later:** automatic tracking, wearable imports.

6.4 Retrieving available air-quality data

- **What:** Backend retrieves PM2.5 observations or forecasts from a verified provider.
- **Why:** A calculation is only as useful as its input data.
- **Example:** Show PM2.5 42 µg/m³, measured at 08:00, provider name, and nearby station distance if available.
- **MVP:** One validated provider path and explicit error/empty states. **Later:** provider fallback and broader coverage. Never hard-code a value as if live.

6.5 Daily pollution-exposure timeline

- **What:** Timeline of logged activities with a concentration label and data-quality status.
- **Why:** Makes time and place patterns easy to see.
- **Example:** Commute 08:00–09:00: higher estimated concentration; classroom 09:00–13:00: data unavailable or proxy used.
- **MVP:** Timeline and visible gaps. **Later:** map-based route replay.

6.6 Estimated cumulative PM2.5 exposure

- **What:** Show a time-weighted average concentration for tracked intervals and optionally a concentration-time sum, clearly naming the metric.
- **Why:** Lets users compare their own tracked days using consistent rules.
- **Example:** 1 hour at 40 µg/m³ and 3 hours at 20 µg/m³ gives a time-weighted average of 25 µg/m³.
- **MVP:** Time-weighted average plus tracked-time coverage. **Later:** more advanced dose models only with expert review.

6.7 Identify higher-estimate locations and periods

- **What:** Rank logged blocks by estimated concentration and duration, without declaring a health outcome.
- **Why:** Helps users see which blocks drove the displayed estimate.
- **Example:** A 90-minute commute at a higher estimated concentration may contribute more concentration-time than a short 10-minute block.
- **MVP:** Show top blocks and the reason. **Later:** route alternatives and traffic-aware models.

6.8 Personalized recommendations

- **What:** Rule-based suggestions linked to logged time and available air-quality information.
- **Why:** The app should turn information into a practical next step.
- **Example:** 'Your planned run is outdoors during a forecast high-PM2.5 period. Consider moving it to a lower-pollution time if practical.'
- **MVP:** Small reviewed rules with source/condition shown. **Later:** richer personalization. Do not use generative AI to invent health advice.

6.9 Weekly exposure summary

- **What:** Summarize daily estimated averages, tracked hours, data coverage and repeated higher-estimate blocks.
- **Why:** Shows patterns without implying that incomplete days are comparable.
- **Example:** 'You logged 22 of 35 planned hours this week; two days have low data coverage.'
- **MVP:** Seven-day view and clear missing-data markers. **Later:** reminders and export.

6.10 Data freshness, sources and uncertainty

- **What:** Display provider, observation/forecast label, timestamp, units, approximate distance if known, and a quality status.

- **Why:** Users need to judge how much to trust an estimate.
- **Example:** 'Nearby observation, 7 km away, 2 hours old — proxy; local conditions may differ.'
- **MVP:** Data timestamp, provider, status and a plain-language caveat. **Later:** calibrated confidence scoring after validation.

07. Understanding Air Pollution and Exposure

Term	Simple meaning	How the app uses it
AQI	An index that turns pollutant levels into a scale/category using a defined system.	Useful for general communication, not the direct input to the exposure calculation.
PM2.5 concentration	Amount of very small airborne particles per volume of air, commonly µg/m³.	Preferred MVP input when reliable, timestamped values exist.
Time in a condition	How long the user says they stayed at a place or activity.	Weight for the time-weighted estimate.
Personal exposure	The pollution concentration around a person over time; actual inhaled dose also depends on breathing and other factors.	The MVP estimates an exposure-related concentration pattern, not exact inhaled dose.
Measured vs estimated	Measured values come from a monitoring instrument; estimated values use a model, nearby station, forecast or proxy.	Label each input and show gaps.

Why not multiply AQI by time? AQI is an index, not a physical concentration unit. Index values are derived from pollutant concentrations using a specific country's or provider's method. Multiplying AQI by hours does not produce PM2.5 concentration-time and cannot be interpreted as a personal dose.

Time-weighted average example (fictional numbers for teaching only):

Time block	PM2.5 concentration	Duration	Concentration × duration
Commute	40 µg/m³	1 hour	40 µg·h/m³
Indoor class (proxy)	20 µg/m³	3 hours	60 µg·h/m³
Total	—	4 hours	100 µg·h/m³

Step 1: add concentration × duration: 40 × 1 + 20 × 3 = 100 µg·h/m³. Step 2: divide by the total tracked time: 100 ÷ 4 = **25 µg/m³** time-weighted average. The result describes these four example hours only. It is not a full-day average if other hours are missing.

The value is not a medical diagnosis, a measurement of the user's lungs, or an exact amount inhaled. Indoor air can differ from outdoor station readings; breathing rate, ventilation, masks, movement and micro-location also matter. The MVP should not claim to calculate inhaled mass.

08. Data Sources

Provider choice is a validation task, not an assumption. Before building the final data adapter, test the exact coordinates and dates in the chosen city, inspect units and timestamps, review terms/rate limits, and document gaps.

Candidate	Potential use	Limitations / verification before use
OpenAQ API v3	Discover monitoring locations and retrieve pollutant measurements; docs describe sensor, hourly and daily resources.	Coverage is not complete worldwide. API key/authentication and terms apply. Verify Indian station coverage, PM2.5 parameter, timestamps, unit and recent data for each target area.
CPCB / Indian AQI portals	Official Indian monitoring and AQI information; potential primary reference for local stations.	Do not assume a public, stable API exists for the needed use. Find current official data access, permitted reuse, station identifiers, update frequency and units. Use only documented/authorized access.

Candidate	Potential use	Limitations / verification before use
Open-Meteo Air Quality API	Hourly model-based air-quality forecast including PM2.5 and PM10, using forecast products.	Forecast/grid value is not a station observation. Check geographical resolution, model/source, forecast time, terms and attribution. Do not label it as a measurement.
Weather provider (optional)	Weather context may help explain conditions or support future features.	Not needed for the first exposure calculation unless a clear use case exists. Verify provider coverage and terms.

Recommended first experiment: check whether the selected provider returns recent PM2.5 observations near one target city and whether it provides enough timestamps for a day timeline. If not, test a forecast source separately and label it as forecast. Do not silently mix observation and forecast values.

Official documentation links are listed in Section 23. Data availability at any specific address is not confirmed by this document.

09. Application Screens

Screen 1 — Dashboard

Purpose: answer 'What does my tracked day look like?' Show city, date, current available air-quality context, tracked hours, time-weighted estimate, data coverage and two or three key blocks. Provide an obvious 'Log activity' action.

Illustrative wireframe:

```
[ Kanpur • Today ]
PM2.5 context: 38 µg/m³ • observation at 08:00
Tracked: 5.5 / 12 logged hours • Coverage: partial
Estimated tracked-time average: 27 µg/m³
Timeline: Commute [higher estimate] | College [proxy] | Walk [missing]
[Add activity] [View My Exposure]
```

Screen 2 — My Exposure

Purpose: explain the estimate. Show timeline blocks, concentration value, observation/forecast/proxy label, time spent, data timestamp and a tap-to-open explanation. Show missing intervals prominently.

Screen 3 — My Places

Purpose: manage Home, College, Office, Bus Stop or Park. Allow add/edit/delete, manual search or map pin where supported, and show the data source/nearest station used for a place. Do not display an exact station distance unless it is calculated from returned coordinates.

Screen 4 — Recommendations and Reports

Purpose: show practical rule-based suggestions and a seven-day report. Each suggestion should say why it appears and which data supports it. Reports should show tracked hours and completeness next to every comparison.

Navigation can be a simple four-item bottom or side navigation. All four screens must work on a typical laptop and mobile viewport.

10. Recommended Technology Stack

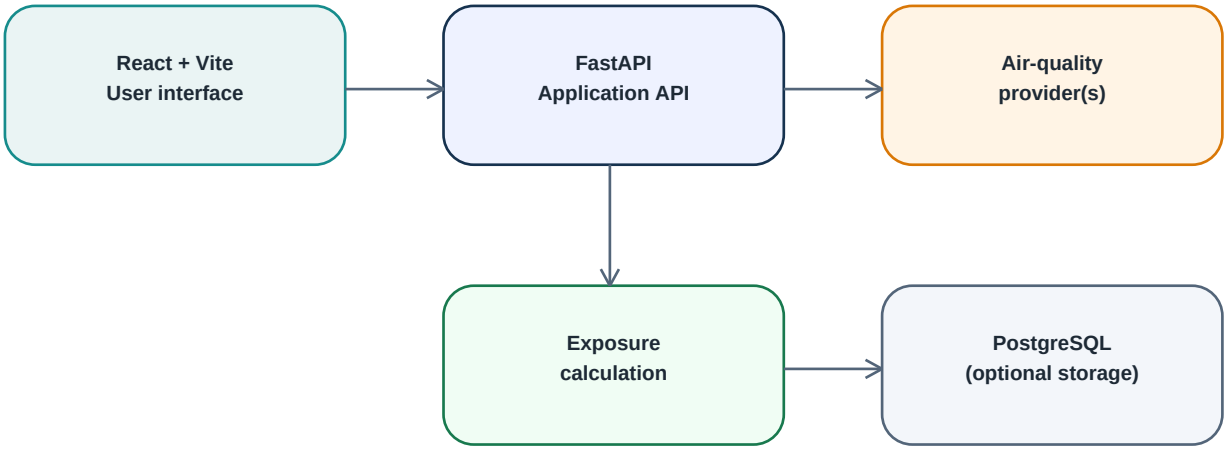
Technology	Role in simple language	MVP recommendation
React + Vite	Builds the interactive screens; Vite runs the local development server and creates production assets.	Use for frontend.
Tailwind CSS	Utility classes for spacing, colors, typography and responsive layouts.	Use to keep styling consistent.
Python + FastAPI	Backend API that validates requests, calls the data provider and runs calculation rules.	Use one small backend service.

Technology	Role in simple language	MVP recommendation
PostgreSQL (optional)	Database for user places, activity logs and summaries that must survive refreshes.	Use if persistent accounts/history are needed; for a demo, local or temporary storage may be enough, but explain the trade-off.
Chart library	Draws a timeline, bars or weekly trend.	Choose one React-friendly library; use a simple chart, not a complex analytics suite.
Map library (optional)	Displays saved places and approximate station locations.	Defer unless a map materially improves the first demo; a typed place list is enough.
AWS hosting	Runs the frontend/backend or stores data, depending on the chosen service.	Choose one deployment path that satisfies the official hackathon rules.

Avoid microservices, machine learning, an AI chatbot, and a complex event pipeline in the first version. They add work without being required to prove the main idea.

11. System Architecture

Keep the system simple. The browser never receives a secret provider API key. It asks the backend for data; the backend calls the external provider, normalizes units and timestamps, runs quality checks and calculation logic, then returns a result.



Concept only: provider access is mediated by the backend; secrets stay server-side.

Data flow:

- 1. User enters places and activity times in the React interface.
- 2. Frontend sends the entries to the FastAPI backend.
- 3. Backend requests observations or forecasts from the chosen provider using server-side credentials if required.
- 4. Backend checks units, timestamps, validity and coverage; it does not replace missing values with fabricated ones.
- 5. Exposure logic computes the estimate for intervals with accepted data and records what was excluded or proxied.
- 6. Backend returns results plus source/quality metadata; the UI displays values, gaps and caveats.
- 7. PostgreSQL stores saved places and activity logs if persistence is enabled.

If the external provider fails, show a clear error or last-known data with its timestamp and stale label. Do not call old data 'current'.

12. Data Models and API Requirements

These are suggested data shapes, not source code. Use IDs to link records. Store timestamps with timezone information and normalize calculation times to a single timezone.

Model	Key fields	Purpose
User (optional for demo)	id, created_at, settings, city	Supports accounts/settings if sign-in is included. Avoid collecting unnecessary profile details.
Place	id, user_id, name, latitude, longitude, place_type, created_at	A saved location. Coordinates should be optional if user types a place that cannot be geocoded.
ActivityLog	id, user_id/session_id, place_id, start_time, end_time, activity_type, notes_optional	Describes where the user says they spent time.
AirQualityObservation	provider, station_or_grid_id, pollutant, value, unit, observed_at, fetched_at, coordinates, data_kind	Keeps the value and provenance needed to explain it.
ExposureSummary	date, tracked_minutes, covered_minutes, time_weighted_pm25, quality_status, calculation_version	Stores a reproducible summary and its coverage.
Recommendation	rule_id, generated_at, message, reason, supporting_data_refs	Lets the app explain why a suggestion appeared.

Suggested API endpoints

Method + path	Purpose
GET /health	Check that the backend is running.
GET /api/air-quality?lat=…&lon;=…	Retrieve available air-quality data for a point; return source, timestamp, units and observation/forecast label.
GET /api/places	List saved places for the current user/session.
POST /api/places	Create a saved place.
PATCH /api/places/{id}	Update a place.
DELETE /api/places/{id}	Delete a place.
GET /api/activities?date=YYYY-MM-DD	List activity blocks for a date.
POST /api/activities	Create an activity block.
PATCH /api/activities/{id} / DELETE /api/activities/{id}	Edit or remove a block.
POST /api/exposure/calculate	Calculate a summary from activity blocks and available data; return coverage and exclusions.
GET /api/reports/weekly?week_start=YYYY-MM-DD	Return daily summaries and tracked-time completeness.
GET /api/recommendations?date=YYYY-MM-DD	Return rule-based suggestions and the reason for each.

For a public demo, a session-based experience can reduce account complexity. If real accounts are added, enforce user-level access checks on every saved record.

13. Exposure Calculation

Recommended MVP metric: the time-weighted average PM2.5 concentration across tracked intervals for which the system has an accepted value. Keep the tracked period explicit. If a concentration c■ applies to duration t■, calculate:

Time-weighted average = Σ(c■ × t■) ÷ Σ(t■)

Only include intervals with accepted input values in the numerator and denominator, and report how many tracked minutes were covered. Do not quietly count missing intervals as zero. If you use an estimated proxy, show it as a proxy

and let the user understand its contribution.

Worked example (fictional):

Block	PM2.5	Hours	Product
Commute	40 µg/m³	1.0	40
Campus	25 µg/m³	4.0	100
Outdoor walk	50 µg/m³	0.5	25
Total	—	5.5	165

165 ÷ 5.5 = **30 µg/m³** time-weighted average for these 5.5 covered hours. It is not the average for the full 24-hour day.

Required data-quality rules

Case	Required behavior
Missing measurement	Mark interval missing. Exclude from estimate; show coverage. Do not use zero.
Stale measurement	Define a freshness threshold with the team. Mark old values as stale; exclude or show as proxy under a documented rule.
Invalid value	Reject impossible/non-numeric values, unknown units or malformed timestamps; log a safe diagnostic.
Different units	Convert only with a known, correct conversion. For mass concentration, normalize to µg/m³. Never confuse AQI with concentration.
Incomplete day	Show tracked time, covered time and gaps. Do not compare a 2-hour partial day with a complete day without a warning.
No nearby station	Show unavailable, or use a clearly labelled model forecast if approved. Do not claim it is a local measurement.
Overlapping logs	Warn and ask the user to resolve, or apply one documented rule. Avoid double-counting time.
Forecast + observation	Keep type and source visible. Prefer separate display or clearly document the combination rule.

The MVP's 'cumulative exposure' label should be used carefully. A concentration-time sum (µg·h/m³) is not the same as inhaled mass. Prefer the time-weighted average plus coverage for the first release.

14. Recommendations and User Safety

Use a small set of transparent rules. A recommendation should be connected to a real observation or forecast, include the time it applies to, and avoid guarantees.

Situation	Example wording	Guardrail
Outdoor exercise	“The forecast for your planned time shows higher PM2.5 than the later time shown. If practical, consider moving the activity or choosing an indoor option.”	Do not promise that a change removes all risk.
Commute	“Your logged commute overlaps a higher-estimate period. If your schedule allows, compare another departure time or route.”	Do not claim an alternate route is cleaner without route-level data.
Indoor air	“Outdoor readings do not tell us the exact air inside your room. If you have a suitable indoor purifier or monitor, use its own guidance.”	Do not assume indoor concentration equals outdoor concentration.
Daily habits	“Check the latest source timestamp before making a plan.”	Do not claim food, supplements or exercise undo pollution effects.
Missing data	“We do not have a reliable value for this block, so it is not included in the estimate.”	Honesty is better than a confident but unsupported suggestion.

Recommendations are informational, not medical advice. Avoid personalized disease-risk scores, diagnosis, medication advice or unsupported claims. When giving health-related guidance, use reputable public-health sources and have wording reviewed.

15. Privacy and Security

- Ask for location permission only when needed. Explain what it is used for and provide a manual alternative.
- Do not run background tracking in the MVP. Manual logs are enough to demonstrate the idea.
- Collect the minimum information needed. A saved place can be named 'Home' without collecting a home address as text.
- Keep provider API keys and database credentials on the server, in environment secrets or the chosen hosting platform's secret manager. Never commit secrets to Git or ship them in frontend code.
- If accounts exist, require authentication and verify ownership for every read/update/delete operation.
- Use HTTPS in deployment. Validate inputs and limit request sizes/rates where practical.
- Provide a way to delete saved places, activity logs and account data. Explain what is deleted and whether any backups remain.
- Use fictional/demo data for public presentations unless participants have clearly agreed to share their data.
- Do not expose exact user places in public logs, screenshots, analytics or error reports.

16. MVP Versus Future Features

Build now (MVP)	After MVP proves useful	Do not build yet
City selection; manual saved places; manual activity blocks; one verified data provider; source/timestamp/units; time-weighted PM2.5 estimate; data coverage; daily timeline; basic rules; weekly summary; honest missing-data states; basic tests; deployable demo.	Optional accounts and sync; more cities/providers; notifications; pollution-aware route comparison after suitable route data exists; indoor monitor integrations; data export; more detailed personalization; accessibility refinements based on testing.	Medical diagnosis or personal disease-risk scores; claims of exact inhaled dose without suitable validated sensors/model; AI-generated health advice; always-on GPS tracking; multi-service architecture; complex ML; unverified provider scraping; fabricated live values.

Defer features that depend on data the team has not verified or that add privacy, scientific or maintenance risk. A small, credible working product is stronger than a broad prototype that presents guesses as facts.

17. Testing and Validation

Test calculations separately from the interface. Use fixed fictional inputs so expected results are known. Then test the full journey with live provider data, including failures.

Test	Input / action	Expected result
Weighted average	40 for 1 h; 20 for 3 h	25 µg/m³; 4 covered hours.
Missing interval	Two logged hours; one has no value	Only valid interval contributes; coverage reflects the gap.
Stale observation	Timestamp older than chosen freshness threshold	Marked stale; behavior follows documented policy.
Invalid unit	AQI supplied where PM2.5 concentration is expected	Rejected or separately handled; never treated as µg/m³.
Unit conversion	Known concentration with supported alternate mass unit	Converted correctly; original source unit retained in metadata.
Overlapping activities	Two blocks overlap by 30 minutes	User receives validation warning or documented resolution; no silent double-counting.

Test	Input / action	Expected result
Provider timeout	External API fails	Friendly error; no invented live value; retry/last-known data clearly labelled if used.
No nearby data	Coordinates with no suitable provider result	Unavailable/proxy state shown; no fabricated station.
Timezone boundary	Activity crosses midnight	Assigned to the correct local dates and durations.
Full user journey	Add place → log activities → view estimate → open recommendation	All steps work; source, timestamp, coverage and caveat are visible.
Deletion	Delete a place/activity	Deleted item no longer appears in summaries, according to the documented data policy.

Before a demo, capture the provider response and timestamp used for each displayed real value. Keep a clearly labelled demo mode for screenshots or rehearsals. Never show mock values with a 'live' badge.

18. AWS Hackathon Alignment

The exact AWS requirement depends on the event rules. Read the official rulebook and confirm which AWS services, open-source tools, minimum usage or proof are required. This document does not assume that any particular service automatically qualifies.

A simple deployment pattern is: host the static React frontend on an approved AWS static hosting option; run the FastAPI backend on an approved managed application/container service; and use a managed PostgreSQL-compatible database only if persistent storage is required. Another allowed AWS setup may be simpler. Select the smallest option that the team can deploy and explain.

AWS's role should be real and visible: the deployed product runs on the selected AWS service, environment configuration is safe, logs help diagnose failures, and the demo can show the deployed URL and architecture. Do not add AWS services only as decoration.

Cost awareness: use free-tier or low-cost options only after checking current eligibility and limits; set a budget alert where available; stop unused resources; avoid leaving expensive databases or containers running. Prices and free-tier rules change, so verify them in the official AWS console/documentation.

To demonstrate impact and execution, show a real user flow, an honest data source, a reproducible calculation, an actionable recommendation, and exactly which part AWS hosts or supports.

19. Development Roadmap

Phase	Main tasks	Completion criteria
1. Validate data sources	Check target-city coverage; confirm PM2.5, units, timestamps, auth, terms and request limits.	A short data-source decision note with evidence from real API tests and known gaps.
2. Define calculations	Agree metric, time intervals, stale threshold, missing-data policy, timezone and overlap handling; write test examples.	Calculation rules are documented and unit tests pass on fixed examples.
3. Design core screens	Sketch Dashboard, My Exposure, My Places and Recommendations/Reports; include empty/error states.	A tester can explain each screen and find the main action without help.
4. Implement main journey	Build place and activity entry, validation, timeline and local calculation response.	A user can log a day and see a correctly labelled summary using test data.
5. Integrate real data	Connect backend to selected provider; normalize response and expose provenance/quality.	At least one real, timestamped PM2.5 result works; missing/failed requests are handled.
6. Test and deploy	Run unit and end-to-end tests; secure secrets; deploy frontend/backend; test from another device.	Working deployed URL; no secret in frontend; no fabricated values; main journey passes.

Phase	Main tasks	Completion criteria
7. Prepare three-minute demo	Prepare one story, a test account/demo routine, source evidence, calculation example and AWS explanation.	Demo fits the time and includes a fallback if live provider access fails.

No fixed deadline is assumed. Estimate effort after data access and hackathon rules are confirmed; the data-source validation is an early gate because the product depends on credible input.

20. Definition of Done

The MVP is ready only when the team can check off the conditions below:

- [ ] A new user can select a city without granting location permission.
- [ ] User can add, edit and delete places.
- [ ] User can log, edit and delete activities with valid times.
- [ ] The app handles overlapping blocks and timezone boundaries by a documented rule.
- [ ] The selected data provider is verified for at least one tested target area, or the UI clearly says data is unavailable.
- [ ] Every displayed air-quality value has a source, unit, timestamp and observation/forecast label.
- [ ] PM2.5 values are distinguished from AQI.
- [ ] The time-weighted average matches the hand-calculated test examples.
- [ ] Missing and stale values are visible; no missing value is replaced by zero or fabricated data.
- [ ] The daily timeline explains which blocks are covered and which are not.
- [ ] At least one recommendation is triggered by a transparent rule and explains why it appears.
- [ ] Weekly summary shows tracked hours and coverage, not only a score.
- [ ] The interface clearly states that estimates are not exact personal measurements or medical advice.
- [ ] Provider secrets are not exposed in frontend code or source control.
- [ ] User data can be deleted according to the documented behavior.
- [ ] Main journey and failure cases are tested.
- [ ] The deployed version works and AWS/event-rule compliance has been verified.
- [ ] A three-minute demo and fallback plan are ready.

21. Three-Minute Hackathon Demo

Time	What to show	What to say
0:00–0:25	Problem and product statement	“A city AQI does not describe every person's daily routine. This app connects logged time and place with available air-quality information.”
0:25–0:55	Choose city; open saved places; add a short activity	Show how a student logs a commute and college time. Mention manual entry keeps tracking optional.
0:55–1:35	Open My Exposure timeline	Point to real source, timestamp, PM2.5 unit, data coverage and missing intervals.
1:35–2:05	Show calculation	Use the fixed example: (40×1 + 20×3) ÷ 4 = 25 µg/m³. Say the numbers are illustrative unless they came from the displayed provider response.
2:05–2:35	Show recommendation and weekly summary	Explain the rule and show tracked-time coverage so the app does not overstate incomplete data.
2:35–3:00	Show deployed app and AWS contribution	Name the exact AWS service used, what it hosts, and the verified event-rule requirement it satisfies.

Demo safety net: prepare a labelled sample dataset and screenshots for network failure. Clearly say 'illustrative demo data' when using it. Never label sample data as live measurements.

22. Risks and Important Decisions

Risk / decision	Why it matters	Action before or during build
Local data coverage	A provider may not have recent PM2.5 data near every place.	Test target-city coordinates before choosing the main data path.
Station-to-person mismatch	A station reading may differ from a street or indoor room.	Show approximate source area and caveat; do not call it personal measurement.
Forecast mistaken for observation	Forecasts are model outputs, not observed station readings.	Store and display data_kind separately.
Misleading metric	AQI × time is not a physical exposure calculation.	Use concentration values with compatible units and documented time weighting.
Incomplete logs	Users may forget to log parts of the day.	Show tracked and covered time; avoid false comparisons.
Privacy / sensitive places	Home, college and commute reveal routine.	Manual mode, minimal storage, access control and deletion.
API dependency	Provider downtime, auth changes or rate limits can break results.	Handle failures, monitor errors and prepare a clearly labelled demo fallback.
Scope creep	Maps, GPS, AI and many providers can consume the build time.	Protect the core journey; add only after the MVP works.
AWS rule mismatch	A technically deployed app may not satisfy event requirements.	Verify exact allowed services, open-source conditions, proof and submission rules.
Recommendation safety	Poor wording may imply medical certainty.	Use reviewed rules, show reasons and avoid diagnoses or guaranteed outcomes.

Decisions to settle before development:

- Which single city and which exact provider will be used for the first working demo?
- Will the MVP use observations only, forecasts only, or separate both? Avoid blending them without a documented rule.
- What freshness threshold will define a stale value, and how will nearby stations be selected?
- Will users need accounts, or can a session-based demo prove the idea?
- Which AWS services are explicitly accepted by the hackathon, and what proof must be submitted?
- What is the simplest map/place-entry approach that is reliable in the available time?

23. References

The links below are official or primary documentation pages consulted for this specification. Re-check them before implementation because API details, access requirements, terms and pricing can change.

Air-quality and health context

1. World Health Organization — Global Air Quality Guidelines (2021): https://www.who.int/publications/i/item/9789240034228

OpenAQ

2. OpenAQ API overview and authentication: https://docs.openaq.org/api

3. OpenAQ API information and coverage caveat: https://docs.openaq.org/about/about

4. OpenAQ measurements and aggregation resources: https://docs.openaq.org/resources/measurements

5. OpenAQ locations API documentation: https://docs.openaq.org/api/operations/locations_get_v3_locations_get

Forecast data

6. Open-Meteo Air Quality API documentation: https://open-meteo.com/en/docs/air-quality-api

India-specific source to investigate

7. Central Pollution Control Board (CPCB), Government of India: https://cpcb.nic.in/. The team must verify current official data access, terms and whether a documented interface supports the target use case.

Technology documentation

8. React: https://react.dev/ • Vite: https://vite.dev/guide/

9. Tailwind CSS: https://tailwindcss.com/docs • FastAPI: https://fastapi.tiangolo.com/

10. PostgreSQL documentation: https://www.postgresql.org/docs/

11. AWS documentation and service pricing: https://docs.aws.amazon.com/ • https://aws.amazon.com/pricing/

Accuracy note: This specification does not confirm real-time data availability for any specific address or station, does not guarantee CPCB API access, and does not determine which AWS services qualify under the event rules. Those are explicit validation tasks for the team.

Final MVP checklist: use Section 20 as the release gate. If a feature cannot be supported by reliable data or tested behavior, show it as unavailable or defer it rather than filling the gap with a guess.